

Proof. **Bernoulli's Inequality.**

Statement. Let $x \in \mathbb{R}$ with $x \geq -1$ and let $n \in \mathbb{Z}$ with $n \geq 0$. Then

$$(1+x)^n \geq 1+nx.$$

1. Degenerate cases.

- $n = 0$: $(1+x)^0 = 1 = 1 + 0 \cdot x$.
- $x = 0$: $(1+0)^n = 1 = 1 + n \cdot 0$ for every n .
- $x = -1$: $1+x = 0$ and hence $(1+x)^n = 0$. The right-hand side equals $1-n$. Because $n \geq 0$, $0 \geq 1-n$; equality occurs only when $n = 1$.

These three situations also appear in the final description of equality.

2. Induction on the exponent. We prove the claim for all integers $n \geq 0$ by induction.

Base step $n = 1$.

$$(1+x)^1 = 1+x = 1+1 \cdot x,$$

so the inequality holds with equality.

Inductive hypothesis. Assume that for a fixed integer $k \geq 1$ and for every real $x \geq -1$,

$$(1+x)^k \geq 1+kx. \tag{1}$$

Inductive step $k \rightarrow k+1$. Since $x \geq -1$ we have $1+x \geq 0$. Multiplying (??) by this non-negative factor preserves the inequality:

$$(1+x)^{k+1} = (1+x)^k(1+x) \geq (1+kx)(1+x).$$

Expanding the right-hand side gives

$$(1+kx)(1+x) = 1 + (k+1)x + kx^2.$$

Because $k \geq 1$ and $x^2 \geq 0$, the term kx^2 is non-negative. Dropping this non-negative term yields

$$(1+x)^{k+1} \geq 1 + (k+1)x.$$

Thus the statement is true for $k+1$. By the principle of mathematical induction,

$$(1+x)^n \geq 1+nx \quad \text{for all } n \geq 0 \text{ and } x \geq -1.$$

3. Equality cases. In the inductive step equality can occur only if

1. equality holds in the inductive hypothesis $(1+x)^k = 1+kx$; and
2. the additional term kx^2 vanishes.

Since $k \geq 1$, condition 2 forces $x^2 = 0$, i.e. $x = 0$. Tracing back through the induction, equality is therefore possible exactly in the following situations:

- $n = 0$ (the trivial exponent);
- $n = 1$ (the base case, where the formula is an identity for any admissible x);
- $x = 0$ (any exponent).

The boundary value $x = -1$ yields equality only when $n = 1$, which is already covered. Hence the complete equality condition is

$$\boxed{n = 0 \text{ or } n = 1 \text{ or } x = 0}.$$

4. Remarks on a direct proof via the binomial theorem. Writing the binomial expansion,

$$(1+x)^n = 1 + nx + \sum_{j=2}^n \binom{n}{j} x^j = 1 + nx + x^2 \sum_{j=0}^{n-2} \binom{n}{j+2} x^j,$$

the factor x^2 is non-negative for $x \geq -1$, and every coefficient $\binom{n}{j+2}$ is non-negative. Consequently the whole tail

$$x^2 \sum_{j=0}^{n-2} \binom{n}{j+2} x^j \geq 0,$$

which leads again to $(1+x)^n \geq 1+nx$.

Conclusion. For every real number $x \geq -1$ and every integer $n \geq 0$,

$$(1+x)^n \geq 1+nx,$$

with equality precisely when $n = 0$, $n = 1$, or $x = 0$. □ □